Arista Access Node QoS/Congestion Event Troubleshooting and Resolution Procedure

Node: MAR-ACC-36 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_2130

Description

The failure domain of QoS/Congestion Events, specifically microbursts, interface tail drops, and QoS queue starvation, on the MAR-ACC-36 node at the Marathahalli Core site, necessitates a comprehensive understanding of the underlying protocol and hardware mechanics. This Access role, leveraging Arista equipment, is susceptible to such failures due to the intricacies of packet switching and queuing mechanisms. The Arista EOS (Extensible Operating System) manages traffic flows through a combination of hardware and software components. However, during periods of high network utilization or when traffic patterns exhibit bursty characteristics, the finite resources of the network interfaces and the queues can become overwhelmed, leading to congestion and subsequent packet drops. This scenario is exacerbated by the lack of sufficient QoS policies that prioritize critical traffic, thereby ensuring that all traffic types are treated equally and potentially leading to starvation of high-priority queues. The critical nature of this failure, as indicated by the ALARM_2130, underscores the need for immediate and precise diagnostic and resolution actions to mitigate the impact on network performance and user experience.

Pre-requisite Checks

- Verify the physical integrity of the MAR-ACC-36 node and its connections at the Marathahalli Core site.

- Confirm the status of redundant links to ensure no single points of failure exist.

- Check for any recent configuration changes or software updates that could be contributing to the issue.

- Ensure that remote hands are available at the site for potential hardware interventions.

Potential Root Causes

Cause	Probability
Inadequate QoS Configuration	High
Insufficient Bandwidth Allocation	Medium
Hardware Failure (e.g., ASIC, Interface Card)	Low

Diagnostic Commands & Expected Outputs

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Command: show qos interface eth 1/1, Expected Output: Displays the current QoS configuration and statistics for the specified interface., Description: This command isolates the QoS configuration applied to a specific interface, allowing for the assessment of whether the current
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policies are adequate for the traffic types and volumes being handled.

Command: show interfaces counters ethics, Expected Output: Provides detailed interface counters, including error and drop statistics., Description: Analyzing the interface counters helps in identifying if there are any physical or data link layer issues contributing to the congestion or packet drops.

Command: show running-config section qos, Expected Output: Displays the current QoS configuration applied to the device., Description: Reviewing the QoS configuration section of the running-config helps in understanding the priority queues, policing, and shaping policies that are in effect, which can contribute to or alleviate congestion events.
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Resolution Steps

Step 1: Step: Apply Temporary QoS Policy, Command: qos policy-group add 10G-POLICY interface eth 1/1, Description: Applying a temporary QoS policy to prioritize critical traffic and mitigate congestion. This step requires careful consideration of the existing traffic patterns and the specific needs of the network.

Step 2: Step: Increase Bandwidth Allocation, Command: interface eth 1/1 bandwidth 10G, Description: Increasing the bandwidth allocation on the affected interface can help in reducing congestion, assuming the physical link can support the increased rate and there are no upstream bottlenecks.

Step 3: Step: Disable and Re-enable Affected Interface, Command: shutdown interface eth 1/1; no shutdown interface eth 1/1, Description: In some cases, simply disabling and then re-enabling the affected interface can reset the hardware and software components, potentially clearing the condition causing the congestion event.

Post-Check Verification

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Command: show qos interface eth 1/1, Expected Output: The QoS configuration and statistics should reflect the changes made during the resolution steps., Description: Verifying the QoS configuration post-change ensures that the policies are correctly applied and are having the desired effect on traffic management.

Command: show interfaces counters ethics, Expected Output: Error and drop statistics should decrease or stabilize post-resolution., Description: Monitoring the interface counters after applying the fixes helps in confirming whether the congestion event has been successfully mitigated.
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Roll-back Procedures

- Step: Remove Temporary QoS Policy, Command: no qos policy-group 10G-POLICY interface eth 1/1, Description: If the applied QoS policy does not yield the expected results or introduces unintended consequences, it should be removed to restore the original configuration.

- Step: Revert Bandwidth Allocation, Command: interface eth 1/1 no bandwidth 10G, Description: If increasing the bandwidth allocation does not resolve the issue or causes other problems, the original bandwidth setting should be restored.